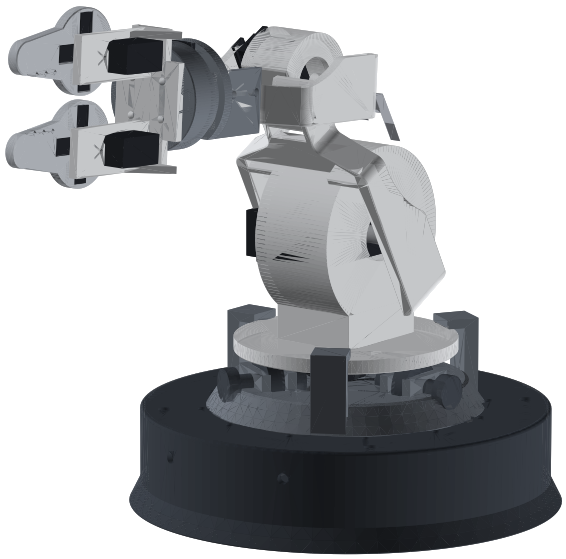
Robot Arm

5-axis desktop arm · ESP32 · 3D printed · build manual



Printed parts	**11 (+2 optional)**
Print plates	**6**
Filament	**≈430 g PETG**
Reach, shoulder to wrist	**150 mm**
Rated payload	**40 g (46 g limit, set by the elbow)**
Base	**Ø186 × 42 mm**

OVERVIEW

How the arm works

Five axes

Base yaw turns the whole arm about a vertical axis. Shoulder and elbow lift and fold it. The wrist ROLLS — it spins the hand about the arm's own axis — and the last two servos are the claw, one per finger. Every joint is a hobby servo driving a printed part directly: no gearboxes, no belts and no gears anywhere. There ARE bearings, five 608ZZ, but they carry load beside the servos rather than transmitting anything.

The turntable is the clever bit

The base does NOT hang the arm off a servo shaft. Three 608ZZ bearings run on horizontal axles inside the turret cap and bear on the flat underside of the shoulder bracket's disc. They take the arm's weight and its overturning moment, which leaves the base servo transmitting torque only. That is what lets the smallest servo in the kit drive the base: gravity does no work about a vertical axis, so once the bearings carry the tipping load there is almost nothing left to hold.

Why the joints are drums

Each joint is a cylinder concentric with its own axis. The part holding a servo's CASE carries the drum; the part driven by its HORN carries a skirt that laps over it. Both being concentric means the joint stays closed at every angle and neither part sweeps into the other. It also means each drum's far cap can carry an idler bore, so every joint is in double shear rather than hanging off the servo's output shaft alone.

What limits it

Not the plastic — bending stress in the loaded sections is about 0.02 MPa against a conservative 12 MPa allowable, a margin of several hundred times. The limit is servo duty. Held straight out, the elbow is the busiest joint at 46 % of its stall torque, and that is what sets the 40 g payload.

BILL OF MATERIALS

What you need to buy

ITEM	QTY	NOTES
MG995 servo	1	Shoulder — the only joint gravity really loads
MG90S servo	2	Elbow and wrist pitch
SG90 servo	3 (+1 spare)	Base yaw, and one per claw finger
608ZZ bearing	5	3 turntable rollers + 2 idler journals
M8 × 30 bolt + nyloc nut	3	Roller axles
M3 heat-set insert, Ø4.0 hole	10	7 in the base, 3 radial in the cap
M3 × 10 socket screw	10	Turret, lid, clips
M3 heat-set insert (rotator)	4	Effector bolt circle, Ø22
M3 × 8 socket screw	4	Hand or scissors onto the rotator
M3 × 30 socket screw + nyloc	2	wrist_mount through the forearm
M3 × 10 self-tapper	2	Upper arm cover
ESP32 dev board	1	38-pin DevKit assumed
400-point breadboard	1	85.5 × 55.5 measured, bay is built to it
5 V 5 A PSU + DC lead	1	See the power page — 5 A is not optional
Toggle switch, Ø13.1 thread	1	Rim, switches the whole supply
5 mm LED + 220 Ω resistor	2	Rim, press fit — now on GPIO 32/2
1000 µF electrolytic, 10 V+	1	Across the servo rail
Zip ties, 2.5 mm	~6	Cable strain relief
Servo screws	—	Use the ones supplied with each servo

Everything else is printed. No glue is used anywhere;
the arm comes apart again.

PRINTING

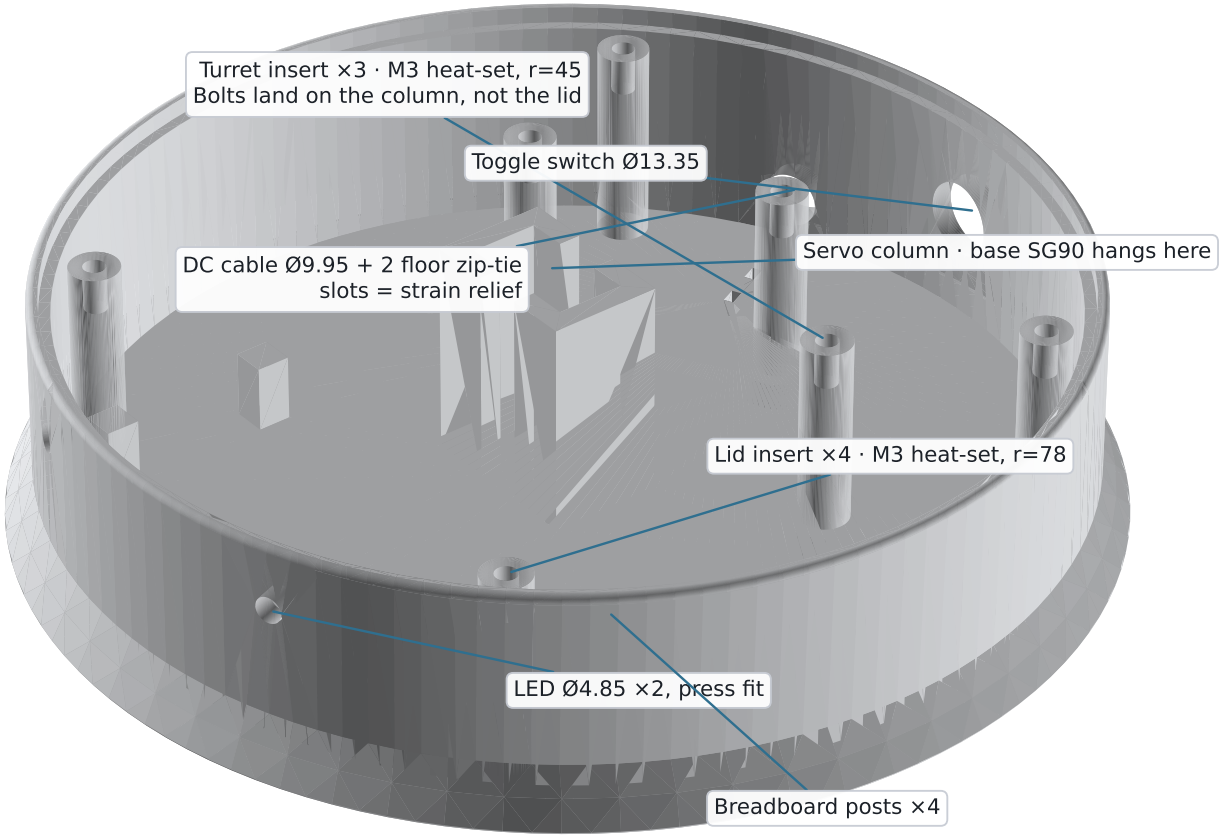
Print plates

FILE	CONTAINS	ORIENTATION	INFILL
p1_base	base_platter	flat, bay up	25 %, 4 walls
p2_lid	base_lid	flat, either face	15 %, 3 walls
p3_turret	turret_cap, shoulder_bracket	disc down, drum up	40 %, 4 walls
p4_links	upper_arm, forearm, cover	flat, channel up	30 %, 4 walls
p5_hand	bracket, rotator, claw ×5	as modelled	40 %, 4 walls
p6_scissors	scissor_cradle, effector_lever	flat	30 % — INCOMPLETE, see note

PETG throughout: tougher than PLA at the joints and it does not creep in a warm room. Bed 70–80 °C, no chamber heat needed. NO PART NEEDS SUPPORT MATERIAL — if your slicer wants to add any, the part is oriented wrong.
Print p3_turret first: it holds the bearings and is the least forgiving plate.
DO NOT PRINT p6 YET. The cradle pins into the wrist rotator correctly, but the effector servo lost its mount when the wrist was rebuilt, and re-siting it depends on the real scissors — every scissor dimension in the model is still an assumption. Measure yours first; see specs/robotarm/scissor_cradle.md.

PART REFERENCE

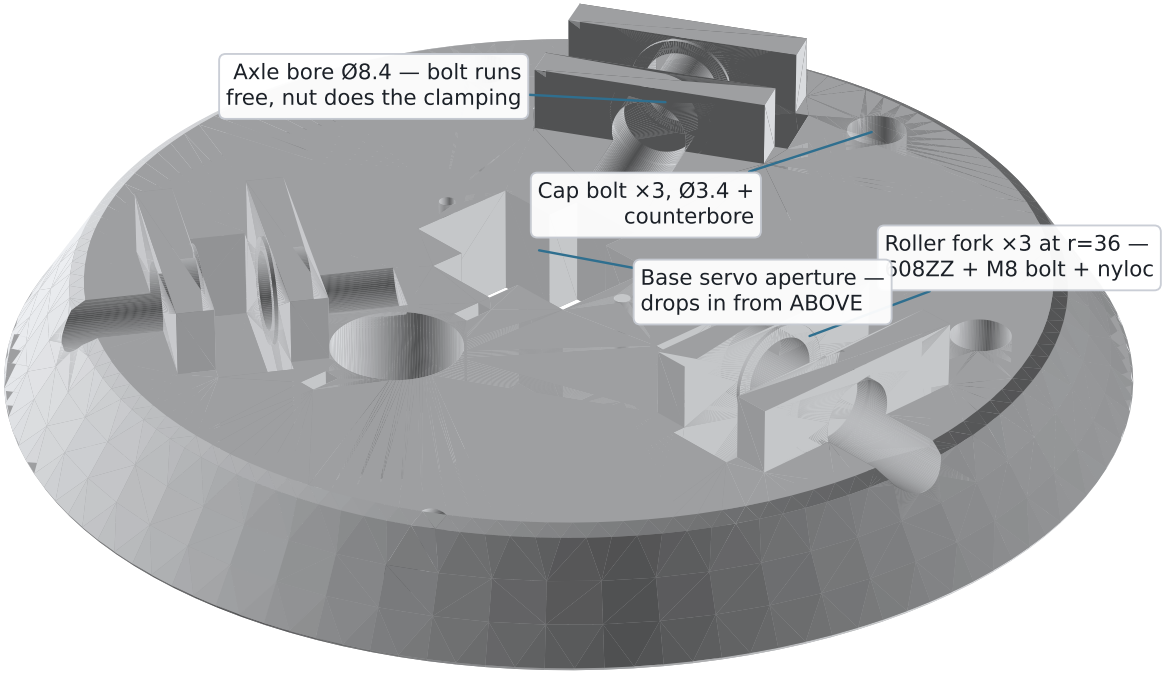
Base platter



Heat-set all seven inserts FIRST, while the bay is empty. The three at r=45 are the ones the turret bolts to; they pass up through the lid so the load lands on the column. The breadboard needs no fixing — the column wall stops it inboard and the four posts hold its sides.

PART REFERENCE

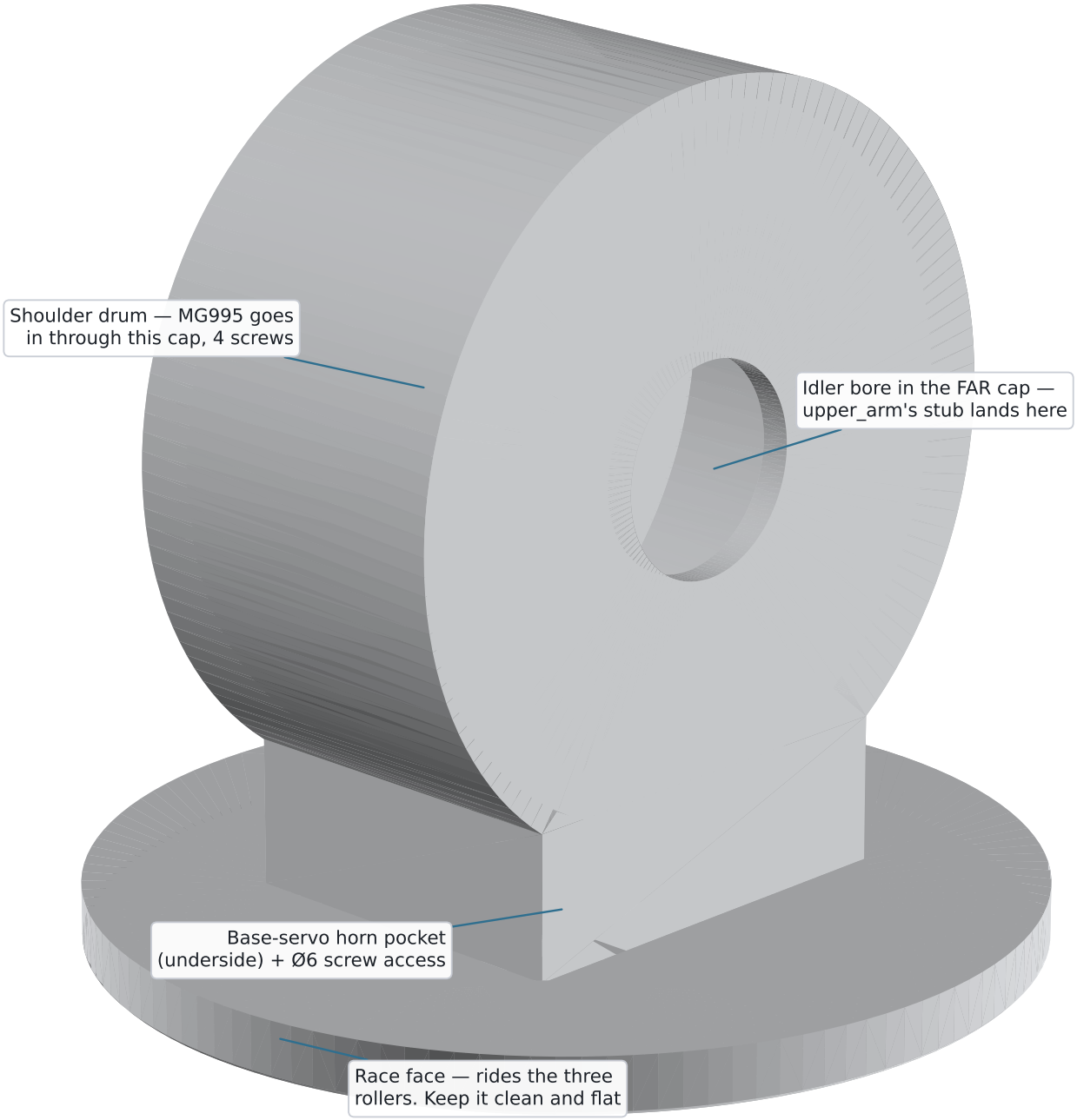
Turret cap



The clamp pads inside each fork touch the bearing's INNER race only, so the outer race never rubs. After assembly every bearing must still spin freely by hand — if one is stiff, the nyloc is over-tightened or a pad printed proud.

PART REFERENCE

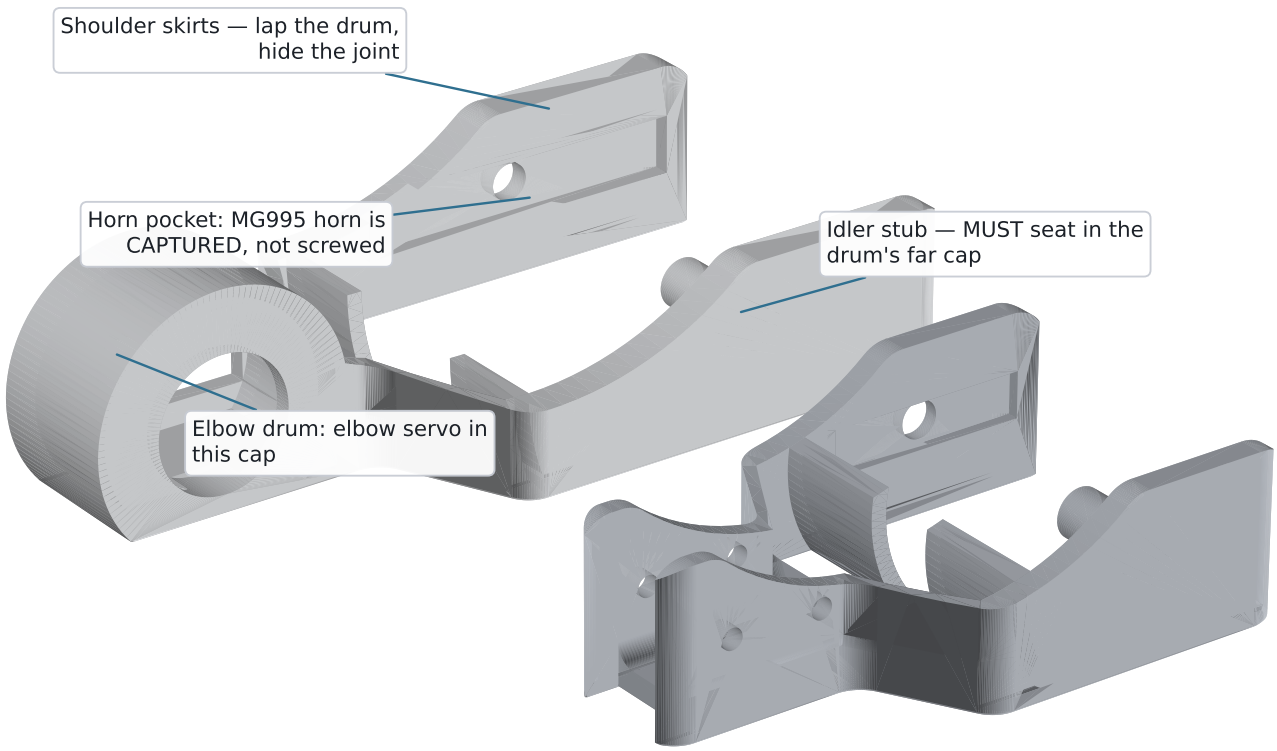
Shoulder bracket



This part is both the turntable and the shoulder housing. The disc's flat underside is a bearing race: do not sand it, and do not print this part on its side — the race has to come off the bed flat.

PART REFERENCE

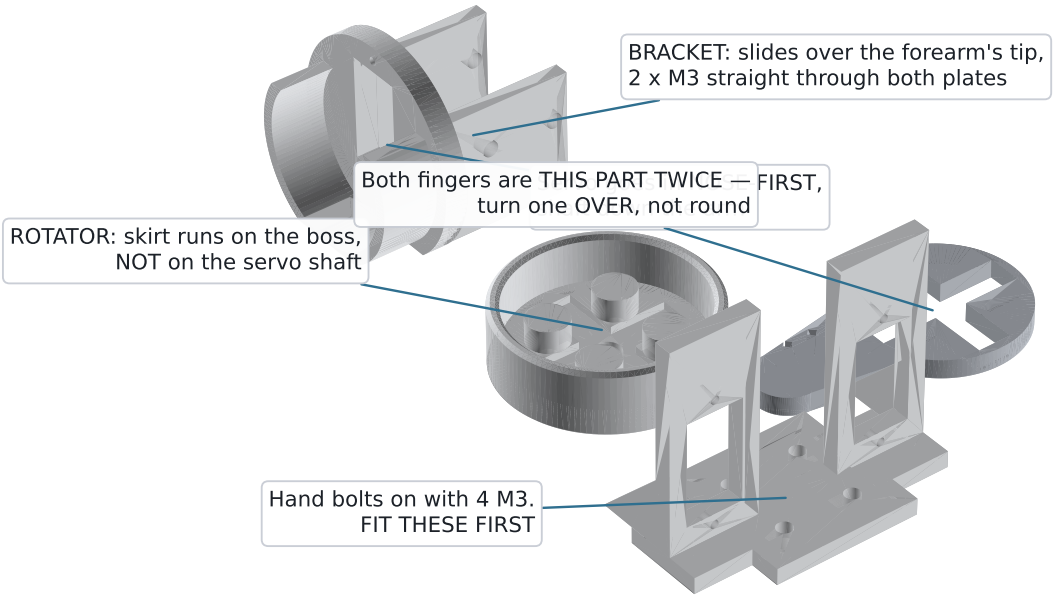
Upper arm and forearm



Both links are open channels on top so the servo can be fitted and its lead routed; the gable cover closes the upper arm afterwards. The channel must stay clear near the shoulder — that space is swept by the bracket as the arm lifts.

PART REFERENCE

Wrist and hand



The wrist ROLLS, and it bolts on. wrist_mount is a saddle that slips over the forearm's tip and takes two M3 straight through it and both of the link's side plates; the servo sits in it nose-first with its shaft pointing forward down the arm. The effector's weight is carried by the rotator's SKIRT running on the boss in front of that bracket, not by the servo's 4.8 mm shaft.

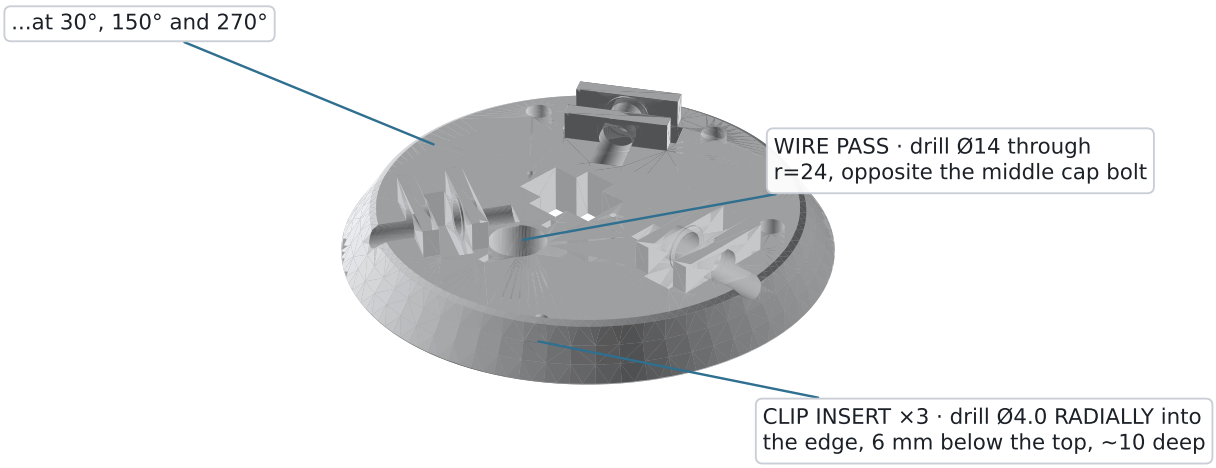
The hand is a claw with ONE SERVO PER FINGER — no gears, no rack, no linkage. Each finger bolts straight to its own horn. It opens 0-43 mm with about 0.25 kgf per finger.

BOTH FINGERS ARE THE SAME PRINTED PART, and the second one goes on TURNED OVER rather than turned round: flipping it is what swaps which edge grips. That is the reason its horn pocket is cut clean through, and the reason the horn's own screw needs a WASHER under the head.

What it costs, plainly: the fingers rotate, so their faces only meet flush at one opening and roll off anything wider or narrower.

RETROFIT

Already printed an earlier version?



REPRINT the turret cap. Earlier plates had the three roller forks turned 90 degrees, with the axles laid tangentially. That is wrong and cannot be drilled out: the disc's surface moves tangentially over each roller, so a tangential axle makes the bearing scrub instead of roll and wear a flat on its outer race. The axle has to point at the hub. Nothing else on plate 3 changed, so reprint just this part.

Everything else is a drill or a reprint of one small item:

shoulder_bracket	open the centre bore from Ø6 to Ø14, right through. It is the horn-screw access AND the wire pass.
print	turntable_clip x3, about 4.7 g each
plates 1, 2, 4	unchanged — keep what you have

THE FOREARM AND ALL OF PLATE 5 ARE NEW. The wrist ROLLS now instead of pitching, and it bolts on: the forearm's far end is a plain square-ended channel with two cross-drilled M3, so the link is a different part and has to be reprinted. The upper arm and its cover are unchanged — print the forearm on its own. Plate 5 is all new too: a bracket, a rotator, a claw body and two fingers.

PLATE 6 IS ON HOLD. The cradle bolts to the rotator like the hand does, but the effector servo lost its mount with the old wrist and re-siting it needs the scissors measured. Do not print it yet.

Servo pockets on earlier plates have no notch for the servo's own lead, so the servo will not push home with its wire attached. Current plates have the notch printed in; on an older part, file a 7 x 4 mm slot in the side of the pocket where the lead comes out. See the fitting page.

FITTING

Which holes to open, and which to leave

FEATURE	SIZE	FIT	IF IT IS TIGHT
M3 heat-set insert	Ø4.0	interference	LEAVE IT. The insert needs the grip
5 mm LED	Ø4.85	press	LEAVE IT. Push harder, or warm the LED
M3 screw clearance	Ø3.4	clearance	Open to 3.5 if a screw binds
M8 roller axle	Ø8.4	loose	Already loose; the nut does the clamping
Toggle switch	Ø13.35	sliding	Open to 13.6 — thread tolerances vary
DC cable gland	Ø9.95	sliding	Open as needed
Wire pass, cap	Ø14.0	generous	Open to 16 if 4 connectors will not pass
Turntable centre bore	Ø14.0	generous	Open to 16, but stay under 18 or the horn pocket goes
Servo case pocket	case + 0.25	sliding	File the LEAD NOTCH, do not widen the pocket
Servo lead notch	7 × 3.5	clearance	Deepen freely — it carries nothing

PETG holes print UNDERSIZE, typically 0.1–0.3 mm, and the first layer is tighter again because of elephant's foot. That is why several of these are meant to be cleaned up on assembly rather than printed to size.

HOW: run the drill SLOWLY. PETG melts and the flutes grab, which snatches the part and tears the hole oval. Step drills are much safer than twist drills in thin plastic. For pockets use a flat file, not a drill — you want to relieve one face, not enlarge the bore. Go one size under, test the fit, then creep up.

Assembly · base and turret

1 Heat-set the base inserts

Seven M3 inserts into base_platter: three at r=45, four at r=78. Do this before anything else goes in the bay.

2 Prepare the turret cap · RETROFIT ONLY

If your cap came off an earlier plate, drill it NOW while it is loose: the Ø14 wire pass, and three radial Ø4.0 holes in the edge. Heat-set an M3 insert into each of the three. See the retrofit page. Current plates have all of this printed in.

3 Fit the rim electronics

Toggle switch, DC lead and the two LEDs. Zip-tie the DC lead to the floor through the two slots beside its hole — that is the strain relief, and without it every tug lands on the switch terminals.

4 Drop in the breadboard

It locates against the column wall and the four posts.

5 Base servo into the turret cap

SG90 down through the rectangular opening from above, flange screwed to the cap's top face. Its lead sits in the notch in the side of the pocket — if the servo will not seat with the wire on, that notch is what needs filing, not the pocket.

6 Build the three rollers

608ZZ between each fork's pads, M8 × 30 through, nyloc snug. Every bearing must still spin freely afterwards — the pads clamp the INNER race only, so the outer race never rubs.

Assembly · closing the base

7 Lid on, then turret cap

The lid drops over the three bosses; the cap bolts to those bosses with 3 M3 screws and traps the lid. No visible fasteners in the lid itself.

8 Feed the arm loom up through the wire pass

Before the turntable goes on, pull the four servo leads (shoulder, elbow, wrist and the two claw servos) up through the cap's Ø14
hole and zip-tie them to it. Leave a generous slack loop in the 12 mm gap — that loop is what lets the base rotate.

9 Shoulder bracket onto the rollers

Thread the loom up through the disc's Ø14 centre bore FIRST. Then press the base servo's cross horn into the pocket underneath, set the disc on the three rollers, and drive the horn's retaining screw up through that same bore. It should turn smoothly with no rock.

10 Fit the three hold-down clips

One M3 radially into each insert in the cap's edge, tongue hooked over the disc's rim. These stop the turntable lifting off a roller when the arm reaches out — without them that load goes into the base servo's shaft. Check the disc still turns freely by hand with all three fitted.

Assembly · arm and hand

11 Shoulder MG995 into the drum

In through the aperture in the near cap, 4 screws. Route its lead out before you fit anything else.

12 Upper arm onto the shoulder

Driven plate onto the horn, and CHECK THE IDLER STUB IS HOME in the far cap's bore before you tighten the horn screw. If it is not seated, the servo shaft carries the whole bending load on its own — the exact failure the drums exist to stop.

13 Elbow servo, then forearm

MG90S into the upper arm's far drum, 2 screws. Forearm onto its horn, idler stub into the far cap. Same check.

14 Route the leads, then the cover

Run the elbow, wrist and gripper leads along the upper arm's channel, then clip the gable cover on with 2 self-tappers through the flank. That is what it is hiding.

15 Wrist bracket onto the forearm's tip

wrist_mount slides straight over the end of the link — it only fits over the PARALLEL part, so push it home until it stops. Two M3 x 30 through the saddle and both side plates, nylocs on the far side. Then the MG90S goes in nose-first: case behind the face, shaft forward down the arm, two screws through the face into its tabs.

16 Rotator onto the horn

Heat-set its four M3 inserts first, from the FLAT face — they bottom out in bosses on the back. Then drop the disc on so its skirt slides over the boss in the tip and the horn lands in the pocket, and drive the horn's own screw with a washer under the head through the Ø6 bore. It must spin smoothly with no rock: that skirt, not the servo shaft, is what carries the whole hand.

17 Hand onto the rotator, THEN its servos

Four M3 x 8 through the hand's flange into the rotator's inserts FIRST — a driver cannot reach past a fitted servo case. Then an SG90 into each upright from the front, leads out the side, two screws each.

18 The two fingers

They are the SAME printed part. Fit the first one; fit the second TURNED OVER, not turned round — flipping it is what swaps which edge grips. Its pocket is cut through, so either face meets the horn. Put a WASHER under each horn screw: there is no material above it to bear on.

CENTRE BOTH SERVOS FIRST and fit both fingers at the same angle, or the tips will not meet.

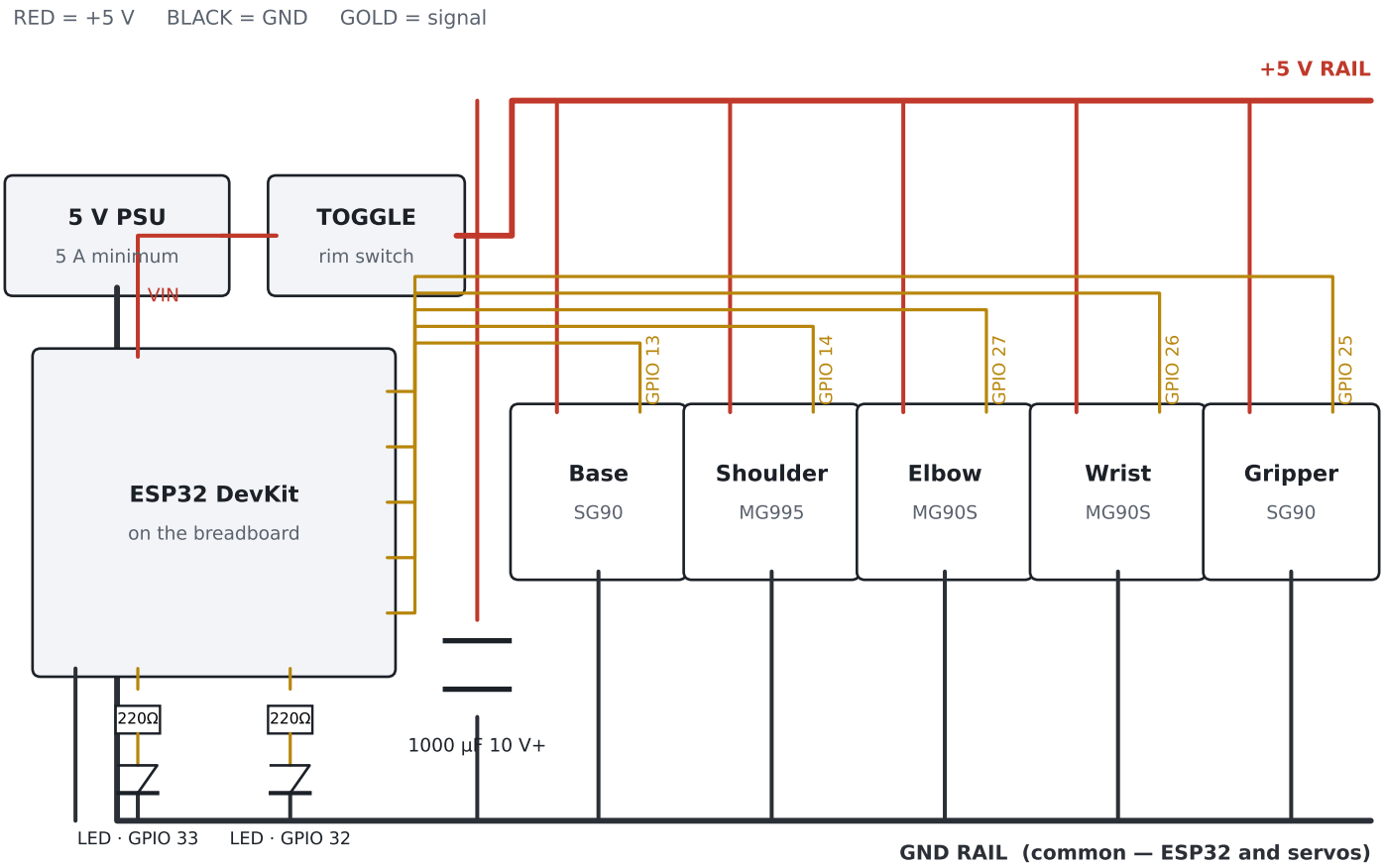
ELECTRONICS

Servo wiring

JOINT	SERVO	SIGNAL PIN	STALL KG·CM	TRAVEL LIMIT
Base yaw	SG90	GPIO 13	1.8	≈180°, software-limited
Shoulder	MG995	GPIO 14	11.0	−95°…+25° from horizontal
Elbow	MG90S	GPIO 27	2.2	−130°…+130°
Wrist pitch	MG90S	GPIO 26	2.2	±90°, set to taste
Hand, left jaw	SG90	GPIO 25	1.8	0…58 mm, 58° of travel
Hand, right jaw	SG90	GPIO 33	1.8	0…58 mm, mirror of left

Signal wires only on the breadboard. Servo POWER must not go through breadboard rails — see the next page. Pins chosen to avoid the ESP32's strapping pins (0, 2, 12, 15), the flash pins (6–11) and the input-only pins (34–39). Drive them at 50 Hz with the LEDC peripheral.

Wiring



Servo POWER (red) comes straight off the rail — NOT through the breadboard's rails, which are good for 1-2 A and will sag under an MG995. Only the ESP32 and the five signal wires (gold) belong on the breadboard. The ground rail is common to everything: the servo signal is referenced to the ESP32's ground, and without that link the servos twitch or ignore you entirely.

Power — read before switching on

Size the supply for stall, not for idle

Five servos stalling together draw roughly 3.9 A: the MG995 alone is about 1.2 A stalled and can spike higher. Use a 5 V supply rated 5 A or more. A phone charger or a USB port will brown out the moment the arm takes a load, and a browning-out ESP32 reboots mid-move.

Do NOT run servo power through the breadboard

Breadboard rails are good for about 1–2 A and their spring contacts have real resistance. The MG995 on its own is beyond what they should carry, and the voltage sag upsets the ESP32 sharing the same rail. Run servo V+ and GND as their own wiring — soldered, or a screw terminal block — and use the breadboard for the ESP32 and the five signal lines only.

Common ground is mandatory

The servo signal is referenced to the ESP32's ground. Tie the servo supply's GND to the ESP32's GND even if they come from different supplies. Without it the servos will twitch, or simply not respond.

One bulk capacitor

1000 µF, 10 V or better, across the servo rail as close to the servos as you can get it, electrolytic polarity observed. It absorbs the current step when a servo starts moving, which is the single most effective thing you can do to stop the ESP32 resetting mid-motion.

What the toggle switch should switch

Put it in the DC positive line ahead of everything, so it kills servo power and the ESP32 together. Switching only the servo rail while the ESP32 stays alive on USB means the arm slams to its commanded pose the instant you switch on.

LEDs

Each 5 mm LED needs its own series resistor — 220 Ω from a 3.3 V GPIO. Wire them GPIO → resistor → LED anode, cathode to GND.

First power-on

Centre the servos BEFORE fitting the horns

This is the step people skip and regret. A servo snaps to whatever position the first PWM pulse commands. With the arm assembled and the horns already fitted at some arbitrary angle, that first command can drive a link straight into the base.

Procedure, one servo at a time, OFF the arm:

1. Wire the servo up on its own.
2. Send a steady 1500 µs pulse at 50 Hz. The servo goes to its centre.
3. With power still on, fit the horn at the angle the assembly needs.
4. Only then bolt that joint together.

Set software end stops

Nothing mechanical stops these joints; the software has to. Use the measured envelopes:

Shoulder	−95° to +25° from horizontal (below that the arm's tail meets the turntable)
Elbow	−130° to +130°
Base	keep clear of the DC lead's exit
Gripper	stop at the closed position; do not drive it into a stall, it will cook the servo

Clamp every command to these in code, not by hand.

First move

Support the arm by hand, power on, and command each joint to its centre one at a time. Watch for: a joint moving the wrong way (swap the direction in software, do not re-fit the horn), a joint buzzing at rest (it is fighting an end stop, or the horn was fitted off-centre), or the ESP32 rebooting (supply or capacitor).

Then check the fingers

Drive both claw servos to their closed end. The two tips should meet on the centreline. If one leads the other, lift that finger off its horn, rotate it a spline tooth and re-seat — the geometry is right in the model, so this is only about where the horn happened to land on its spline.

Set each servo's closed limit where the tips touch and NOT further. Past that point they just push against each other and both servos sit at stall.

The fingers ROTATE, so their faces only meet flush at one opening. That is inherent to a claw, not a fault.

REFERENCE

Load budget and limits

JOINT	SERVO	STALL	HELD	DUTY
Shoulder	MG995	11.0 kg·cm	2.16	20 %
Elbow	MG90S	2.2 kg·cm	1.08	49 %
Wrist roll	MG90S	2.2 kg·cm	0.06	3 %

Arm straight out horizontal, the worst case for every joint, with the parts' real masses and centres of mass. A hobby servo should hold a static load at or under 50 % of stall — stall is a peak rating, not a duty rating. RATED PAYLOAD IS 28 g; the hard limit is 35 g and it is the elbow that sets it, not the shoulder (which would allow 187 g). The wrist barely counts: gravity does almost no work about a ROLL axis, so it only holds whatever you grip off-centre.
Want more? Put the spare MG995 on the ELBOW instead of the MG90S and the ceiling roughly triples. Or shorten the forearm, or lighten anything outboard of it.